

Your Paper Title Here

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Abstract

Your abstract text goes here. Keep it concise, focused on the core database or data management challenge, your approach, and key experimental findings.

CCS Concepts

• Information systems → Data management systems.

Keywords

databases, query processing, distributed systems

1 Introduction

2 Related Work

3 Methodology

3.1 Preliminaries

3.2 Authority and Scope Completeness

Dynamically integrating conjunctive sub-queries during federated query processing in discovery-based querying at runtime is difficult due to the unbounded nature of the query execution. To show this, we first introduce a sufficient condition for correctly delegating conjunctive queries in the traditional federated case. Then we will show how deciding whether we can correctly delegate these queries in the discovery-based querying case is impossible when not all sources have been discovered.

First we define our discovery-based federated query setting in Definition 1 and 2.

Definition 1 (Discovery setting). A *discovery setting* is a tuple $W = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$: a universe of sources $\mathbb{D}$, a map $\text{data} : \mathbb{D} \rightarrow 2^T$ giving the graph each one holds, a set of seeds $\mathcal{D}_0 \subseteq \mathbb{D}$, and a *discovery function* disc assigning to every BGP B a map $\text{disc}_B : \mathbb{D} \times 2^T \rightarrow 2^{\mathbb{D}}$, where $\text{disc}_B(d, \text{data}(d))$ are the sources that a retrieved d exposes while B is evaluated.

Definition 2 (Reachable set, reference answer). Iterating discovery from the seeds, let $\mathcal{D}_{i+1} := \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \text{disc}_B(d, \text{data}(d))$. The *reachable set* is $\text{reach}_B(W) := \bigcup_{i \geq 0} \mathcal{D}_i$, and the *reference answer* to B is $\llbracket B \rrbracket_{G_{\text{reach}_B(W)}}$. Correctness of a delegation is always measured against this reference answer, never against the universe $\mathbb{D}$: sources that are never reached cannot contribute to it.

Given this definition, we recover the traditional federated query processing setting, in which all sources are known beforehand, by taking $\mathcal{D}_0 = \mathbb{D}$ and $\text{disc}_B \equiv \emptyset$.

Next we define an operator to determine the *relevance* of a source for a query execution setting. Relevance records, for each triple pattern and each source, whether that source produces matches. An example of such an operator is the usage of ASK queries to determine exclusive groups in FedX [2].

Definition 3 (Source relevance). Fix a setting W and a BGP B , and write $\mathcal{D} := \text{reach}_B(W)$ for its reachable set; all that follows is

relative to $\mathcal{D}$. Let $\mathcal{P}$ be the set of triple patterns. We write $\llbracket t \rrbracket_d$ for the evaluation of a triple pattern t over the data of source d . The *relevance indicator* $\Delta : \mathcal{P} \times \mathcal{D} \rightarrow \{0, 1\}$ is

$$\Delta(t, d) = \begin{cases} 1, & \text{if } \llbracket t \rrbracket_d \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases}$$

and $\mathcal{D}_t := \{d \in \mathcal{D} \mid \Delta(t, d) = 1\}$ denotes the *relevant sources* of t .

Thus, relevance means that a data source returns answers for a given triple pattern. Next, we introduce a definition of *correct* delegation of conjunctive queries to a set of sources.

Definition 4 (Delegation). Let B be a BGP, $B' \subseteq B$ the delegated fragment, and $S \subseteq \mathcal{D}$ the (only) sources to which B' is sent as a single conjunctive request, where as in Definition 3, $\mathcal{D} = \text{reach}_B(W)$. Write $G_A := \bigcup_{d \in A} \text{data}(d)$ for $A \subseteq \mathcal{D}$. The delegated plan obtains for B'

$$R := \left(\bigcup_{d \in S} \llbracket B' \rrbracket_{\text{data}(d)} \right) \cup \llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}},$$

and the delegation is *correct* if

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}}.$$

This definition follows some of the definitions of federation of FedQPL [1]. Next we give a sufficient condition for correctly delegating conjunctive (sub)-queries to federated sources.

Theorem 1 (Correct delegation). Let W be a federated setting, that is $\text{disc}_B \equiv \emptyset$, so that $\mathcal{D} = \text{reach}_B(W) = \mathcal{D}_0$ is known in advance. Let $t_1, \dots, t_n$ be a set of triple patterns forming a BGP B , $B' \subseteq B$ a sub-BGP, and let $d \in \mathcal{D}$. For d we define $\mathcal{D}' = \mathcal{D} \setminus \{d\}$ the set containing all other sources. Then we can correctly delegate B' to d if $\bigcup_{t_i \in B'} \{d' \in \mathcal{D}' \mid \Delta(t_i, d') = 1\} = \emptyset$

PROOF. We prove the special case of Definition 4 with $S = \{d\}$, so that $\mathcal{D} \setminus S = \mathcal{D}'$. We assume $B' \neq \emptyset$ and that distinct sources use disjoint sets of blank nodes.

The argument rests on the fact that a *single* triple pattern distributes over a union of sources: for every t and every $A \subseteq \mathcal{D}$,

$$\llbracket t \rrbracket_{G_A} = \bigcup_{e \in A} \llbracket t \rrbracket_{\text{data}(e)}, \quad (1)$$

since matching one pattern involves no join. Note that (1) fails for BGPs, which is exactly what makes delegation non-trivial.

By hypothesis no source other than d is relevant to any pattern of B' , so Definition 3 gives

$$\llbracket t' \rrbracket_{\text{data}(d')} = \emptyset \quad \text{for all } t' \in B', d' \in \mathcal{D}'. \quad (2)$$

Step 1. A BGP is the join of its patterns, and each pattern distributes over the sources by (1):

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \bowtie_{t \in B} \llbracket t \rrbracket_{G_{\mathcal{D}}} = \bowtie_{t \in B} \bigcup_{e \in \mathcal{D}} \llbracket t \rrbracket_{\text{data}(e)}.$$

Step 2. Since $\bowtie$ is associative and commutative, the delegated fragment splits off:

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \left(\bowtie_{t' \in B'} \bigcup_{e \in \mathcal{D}} \llbracket t' \rrbracket_{\text{data}(e)} \right) \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}}.$$

Step 3. As $\mathcal{D} = \mathcal{D}' \uplus \{d\}$, the inner union splits, and by (2) its $\mathcal{D}'$ part is empty:

$$\bigcup_{e \in \mathcal{D}} \llbracket t' \rrbracket_{\text{data}(e)} = \underbrace{\bigcup_{d' \in \mathcal{D}'} \llbracket t' \rrbracket_{\text{data}(d')} \cup \llbracket t' \rrbracket_{\text{data}(d)}}_{= \emptyset} = \llbracket t' \rrbracket_{\text{data}(d)}.$$

Substituting into Step 2 and reassembling the join,

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \llbracket B' \rrbracket_{\text{data}(d)} \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}}. \quad (3)$$

Step 4. It remains to identify R from Definition 4. Its second term vanishes, again by (1) and (2) and using $B' \neq \emptyset$:

$$\llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}} = \bigwedge_{t' \in B'} \bigcup_{d' \in \mathcal{D}'} \llbracket t' \rrbracket_{\text{data}(d')} = \bigwedge_{t' \in B'} \emptyset = \emptyset.$$

With $S = \{d\}$ the first term of R is the single request to d , so $R = \llbracket B' \rrbracket_{\text{data}(d)} \cup \emptyset = \llbracket B' \rrbracket_{\text{data}(d)}$.

Conclusion. Substituting R into (3),

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}},$$

which is the correctness condition of Definition 4. $\square$

Theorem 1 gives us a sufficient condition for delegating conjunctive sub-queries in the case where the reachable set $\mathcal{D}$ is known in advance. However, in discovery-based querying the sources are discovered at runtime, governed by some discovery process which returns during the query execution. As such, determining if we can correctly delegate depends on the *state* of the discovery process, which is defined in Definition 5.

Definition 5 (Discovery state). The state of a discovery setting $W = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$ at time T is the tuple

$$W(T) = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc}, \mathcal{D}(T) \rangle,$$

the setting W augmented with $\mathcal{D}(T)$, the set of sources that data has been called on by time T , subject to

$$\mathcal{D}_0 \subseteq \mathcal{D}(T) \subseteq \mathcal{D}(T') \subseteq \text{reach}_B(W) \quad \text{for all } T \leq T',$$

i.e. the seeds are discovered from the start, discovery only ever adds sources, and it never adds one outside what disc can reach. For any given $W(T)$ the discovery *frontier* is

$$F(T) = \left(\bigcup_{d_f \in \mathcal{D}(T)} \text{disc}(d_f, \text{data}(d_f)) \right) \setminus \mathcal{D}(T),$$

the sources that are discovered but do not yet have data called on them. Discovery is complete when $F(T) = \emptyset$.

Next, we introduce the concept of an observation of an engine of the discovery state in Definition 6, which denotes what the engine actually can *know* about its discovery setting at given time T .

Definition 6 (Observationally equivalent states). A discovery state $W(T)$ records the whole world, but a query engine at time T has access only to the *observation*

$$O(T) = \langle \mathcal{D}(T), \text{data}|_{\mathcal{D}(T)} \rangle,$$

the sources it has retrieved together with their contents. Two states $W_1(T)$ and $W_2(T)$ are *observationally equivalent*, written $W_1(T) \sim W_2(T)$, when $O_1(T) = O_2(T)$. Note that $F(T)$ is computable from $O(T)$ and disc: the extent of the frontier is observable, its contents are not. A property of states is *determined at T* when all states

observationally equivalent to $W(T)$ agree on it; an engine at time T can establish a property only if it is determined at T .

Theorem 2 (Correct delegation is undetermined under incomplete discovery). Let $W(T)$ be a discovery state for which $F(T) \neq \emptyset$. Then there exist a BGP B , a sub-BGP $B' \subseteq B$, a set of sources $S \subseteq \mathcal{D}(T)$, and two states $W_1(T)$ and $W_2(T)$, both observationally equivalent to $W(T)$, such that B' is correctly delegated to S in $W_1(T)$ but not in $W_2(T)$. Since the engine's input is $O(T)$ and all three states share it, no procedure can decide whether the delegation is correct.

PROOF. We construct two states $W_1(T)$ and $W_2(T)$, both observationally equivalent to $W(T)$, in which B' is delegated to S correctly and incorrectly respectively. The two agree on $\mathcal{D}_0$, disc and $\mathcal{D}(T)$, and on data over $\mathcal{D}(T)$; they differ only beyond the frontier, which is non-empty by hypothesis.

Take $\mathcal{D}(T) = S = \{d_1\}$ and

$$B' = B = \{(?s, p_1, ?o_1), (?s, p_2, ?o_2)\},$$

$$\text{data}(d_1) = \{(s_1, p_1, o_1), (s_1, p_2, o_2), (s_2, p_1, o_3)\}.$$

Fix some $d^* \in F(T)$, and let every reachable source other than d_1 and d^* hold no data. The two states are then obtained by setting

$$\text{data}(d^*) = \begin{cases} \{(s_3, p_2, o_4)\} & \text{in } W_1(T), \\ \{(s_2, p_2, o_4)\} & \text{in } W_2(T). \end{cases}$$

Since $d^* \notin \mathcal{D}(T)$, both states induce the same observation, and so $W_1(T) \sim W_2(T) \sim W(T)$ holds. The reachable set differs between the two, so we write G_i for $G_{\text{reach}_B(W_i)}$, the union of all data reachable in $W_i(T)$. As only d_1 and d^* hold data,

$$G_1 = \text{data}(d_1) \cup \{(s_3, p_2, o_4)\}, \quad G_2 = \text{data}(d_1) \cup \{(s_2, p_2, o_4)\}.$$

Write

$$\mu_1 = \{?s \mapsto s_1, ?o_1 \mapsto o_1, ?o_2 \mapsto o_2\},$$

$$\mu_2 = \{?s \mapsto s_2, ?o_1 \mapsto o_3, ?o_2 \mapsto o_4\}.$$

Neither added triple matches $(?s, p_1, ?o_1)$, so $\llbracket B' \rrbracket_{\text{data}(d^*)} = \emptyset$ in both states. By Definition 4 we therefore have

$$R_1 = R_2 = \llbracket B' \rrbracket_{\text{data}(d_1)} = \{\mu_1\}.$$

Since $B \setminus B' = \emptyset$ we have $\llbracket B \setminus B' \rrbracket_{G_i} = \{\mu_\emptyset\}$, the join identity, so the correctness condition reduces to $R_i = \llbracket B \rrbracket_{G_i}$.

In $W_1(T)$ the added triple has subject s_3 , which no p_1 -triple mentions, so it contributes no solution and

$$\llbracket B \rrbracket_{G_1} = \{\mu_1\} = R_1 :$$

the delegation is correct. In $W_2(T)$ the added triple joins with $(s_2, p_1, o_3) \in \text{data}(d_1)$ on $?s$, so

$$\llbracket B \rrbracket_{G_2} = \{\mu_1, \mu_2\} \neq R_2 :$$

the delegation is not correct.

The triple at d^* yields no solution on its own, and (s_2, p_1, o_3) yields none within d_1 ; only their join does. A single conjunctive request to d_1 cannot produce it, and neither can one to d^* , so the solution is lost precisely because the join crosses the frontier. As $W_1(T)$ and $W_2(T)$ present the engine with the same observation $O(T)$ while disagreeing on correctness, no procedure reading only $O(T)$ can decide it. $\square$

Lemma 1 (An empty frontier certifies complete discovery). $F(T) = \emptyset$ if and only if $\mathcal{D}(T) = \text{reach}_B(W)$.

PROOF. ($\Rightarrow$) The inclusion $\mathcal{D}(T) \subseteq \text{reach}_B(W)$ holds by Definition 5. For the converse, $F(T) = \emptyset$ says exactly that

$$\text{disc}(d_f, \text{data}(d_f)) \subseteq \mathcal{D}(T) \quad \text{for every } d_f \in \mathcal{D}(T),$$

i.e. that $\mathcal{D}(T)$ is closed under disc; and $\mathcal{D}_0 \subseteq \mathcal{D}(T)$, again by Definition 5. By induction, if $\mathcal{D}_i \subseteq \mathcal{D}(T)$ then

$$\mathcal{D}_{i+1} = \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \text{disc}(d, \text{data}(d)) \subseteq \mathcal{D}(T)$$

by closure, so $\text{reach}_B(W) = \bigcup_{i \geq 0} \mathcal{D}_i \subseteq \mathcal{D}(T)$.

($\Leftarrow$) If $d \in \text{reach}_B(W)$ then $d \in \mathcal{D}_i$ for some i , hence

$$\text{disc}(d, \text{data}(d)) \subseteq \mathcal{D}_{i+1} \subseteq \text{reach}_B(W).$$

Thus $\text{reach}_B(W)$ is closed under disc, and $F(T) = \emptyset$ follows. $\square$

{TODO: Don't like the text here} Lemma 1 is what makes the frontier useful: $F(T) = \emptyset$ is a condition the engine can test, whereas $\mathcal{D}(T) = \text{reach}_B(W)$ is a statement about a set it cannot inspect. Together with Theorem 2 it fixes exactly when delegation can be established at all.

Corollary 1 (Delegation requires complete discovery). The correctness of every delegation of a sub-BGP $B' \subseteq B$ to a set $S \subseteq \mathcal{D}(T)$ is determined at T if and only if $F(T) = \emptyset$.

PROOF. ($\Leftarrow$) Let $F(T) = \emptyset$ and let $W_1(T), W_2(T)$ be observationally equivalent to $W(T)$. As $F(T)$ is computable from $O(T)$, both have an empty frontier, so $\mathcal{D} = \text{reach}_B(W) = \mathcal{D}(T)$ in both by Lemma 1. Then $G_{\mathcal{D}} = \bigcup_{e \in \mathcal{D}(T)} \text{data}(e)$ agrees in both, and since $S \subseteq \mathcal{D}(T) = \mathcal{D}$, so do $G_{\mathcal{D} \setminus S}$ and $\text{data}(d)$ at each $d \in S$. Every term of the condition of Definition 4 therefore agrees, and with it the verdict.

($\Rightarrow$) By contraposition. If $F(T) \neq \emptyset$, then Theorem 2 supplies a B' , an S , and two states observationally equivalent to $W(T)$ that disagree on the correctness of delegating B' to S . That delegation is not determined at T . $\square$

Remark 1. Determination is an information claim, not a computational one. If in addition $\mathcal{D}(T)$ and each $\text{data}(e)$ are finite, correctness is also *decidable* once $F(T) = \emptyset$: the observation fixes $G_{\mathcal{D}}$, so both sides of the condition of Definition 4 can be evaluated and compared. Note that this settles correctness without appeal to Theorem 1, which supplies a cheap sufficient certificate from Δ alone rather than a decision procedure.

From Corollary 1, we find that the problem of runtime federation in discovery-based querying is impossible until all sources are discovered. Depending on the number of sources, which is potentially thousands **{TODO: cite here}**, this is infeasible.

To solve this, we introduce a semantic assumption on what data is considered relevant to our query execution. We call this the *authoritativeness declaration assumption*. In this assumption, we **{TODO: Explain}**. This combines both *authority* who may make statements about a term and *completeness* **{TODO: Explain}**

This assumption is intuitive because **{TODO: Explain intuition with example}**

Formally, we get:

3.3 Delegating Conjunctive Queries

Under this query execution model, we can now investigate delegating queries to composite sources when our model allows determining *correct* delegation under *incomplete* discovery.

3.4 Adaptive Query Planning with Composite Sources

Given the capability of delegating these sub queries, we must now move to the system-side of our problem. Specifically, due to the dynamic discovery of composite sources and the uncertainty of capabilities of these sources, using a static query plan is difficult.

{TODO: Example showing how we can't integrate a sub-query if we have a static plan}

{TODO: Show that even if we use many alternative plans we can still run into issues with overlapping composite sources}

{TODO: Stems!}

3.5 Correctness

3.5.1 Overlapping Sources. Correctness when doing overlapping sources is difficult blabla, cases: Overlapping domain, overlapping triple patterns, both, neither. Given the time dimension, how do we decide what is possible

3.6 Discovery

4 System Application

To evaluate the performance impact of delegating sub-queries we implement hybrid federated link traversal-based query processing. Specifically, within solid blabla

Note this is a concrete application of the stuff introduced here, which is not limited to Solid and link traversal.

5 Evaluation

6 Conclusion

Summarize results and point to future work.

References

- [1] Sijin Cheng and Olaf Hartig. 2020. FedQPL: a language for logical query plans over heterogeneous federations of RDF data sources. In *Proceedings of the 22nd International Conference on Information Integration and Web-based Applications & Services*. 436–445.
- [2] Andreas Schwarte, Peter Haase, Katja Hose, Ralf Schenkel, and Michael Schmidt. 2011. FedX: Optimization Techniques for Federated Query Processing on Linked Data. In *The Semantic Web – ISWC 2011 (LNCS, Vol. 7031)*. Springer, 601–616. doi:10.1007/978-3-642-25073-6_38